

Oral Presentations

Abstract Plenary

OP0001

COMPARATIVE ANALYSIS OF SYNOVIAL TISSUE TOPOGRAPHY BY SINGLE CELL TRANSCRIPTOMICS AND SPATIAL TRANSCRIPTOMICS REVEALS DISCRETE AND OVERLAPPING DISEASE PATHWAYS DRIVING CHECKPOINT INHIBITOR-INDUCED AND SPONTANEOUS INFLAMMATORY ARTHRITIS

Keywords: Cytokines and Chemokines, Oncology, -omics, Synovium, Fibroblasts

J. D. Turner¹, A. K. McLean², A. Thomson³, K. Williams², I. Nicorescu², L. Jones¹, G. Mercedes³, A. Teodosio⁴, B. Hunter³, J. McMurray⁴, S. Gilbert⁴, A. Degnan², D. Scheel-Toellner¹, S. Jariwala⁵, Q. An⁵, R. Plummer⁸, A. P. Croft¹, L. Pallan⁹, B. A. Fisher¹, J. Coxhead², R. Hussain³, M. Magid-Slav⁵, D. McDonald³, S. Siebert⁶, A. Filby³, G. Reynolds¹⁰, A. E. Anderson², C. Hilken², D. F. Tough⁷, A. Filer^{1,4}, A. Pratt¹. ¹University of Birmingham, Rheumatology Research Group, Birmingham, United Kingdom; ²Newcastle University, Translational and Clinical Research Institute, Newcastle upon Tyne, United Kingdom; ³Newcastle University, Flow Cytometry Core Facility, Newcastle upon Tyne, United Kingdom; ⁴University of Birmingham, Birmingham Tissue Analytics, Birmingham, United Kingdom; ⁵GSK, Human Genetics & Genomics, Collegeville, United States of America; ⁶University of Glasgow, School of Infection & Immunity, Glasgow, United Kingdom; ⁷GSK, Respiratory & Immunology Research Unit, Stevenage, United Kingdom; ⁸Newcastle upon Tyne Hospitals, Bobby Robson Unit, Newcastle upon Tyne, United Kingdom; ⁹University Hospitals Birmingham NHS Foundation Trust, Department of Oncology, Birmingham, United Kingdom; ¹⁰Massachusetts General Hospital, Center for Immunology and Inflammatory Diseases, Boston, United States of America

Background: Anti-citrullinated peptide autoantibody seropositive rheumatoid arthritis (SPRA) and psoriatic arthritis (PsA) are relatively well-characterised disease phenotypes for which a growing number of rational therapies target defined pathways. Other forms of seronegative immune-mediated inflammatory arthritis (SIA) – including undifferentiated inflammatory arthritis and seronegative RA – are more heterogeneous in presentation, difficult to classify and lack rational treatment strategies. Immune checkpoint inhibitor-induced inflammatory arthritis (ICI-IA), which develops in ≥6% cancer patients treated with ICIs, shares many clinical characteristics with SIA and is similarly under-represented in interventional clinical trials.

Objectives: Identify common and unique disease mechanisms of SIA and ICI-IA with SPRA and PsA, as potential targetable disease pathways for patients poorly served by existing treatments.

Methods: Synovial tissues were obtained from patients who were naïve to glucocorticoids or disease modifying anti-rheumatic drugs across two recruitment sites. Human biological samples were sourced ethically, and their research use was in accord with the terms of the informed consents under a REC approved protocol. Single-cell RNA-Seq was performed using the 10X platform after enzymatic disaggregation [1] and live cell sorting. Paired paraffin-embedded tissues were haematoxylin and eosin (H+E)-stained and scored for synovitis severity [2]; Xenium spatial transcriptomics is currently underway for a subset of these samples.

Results: Synovitis severity did not differ significantly between disease phenotypes (median Krenn score 5). 138,676 cells were sequenced from 36 individuals (11 SPRA, 7 PsA, 9 SIA and 9 ICI-IA) with major cell lineages clearly delineated following normalisation, clustering and annotation (Figure 1A). Considering relative cellular abundance, a higher proportion of ICI-IA tissues fell within a “fibroblast dominant” (as opposed to myeloid and/or T cell-dominant) cluster, compared with those from participants with “spontaneous” arthritis (45% versus 7%; $p=0.024$, Fisher’s Exact test). In addition, ICI-IA tissues exhibited a significant preponderance of the endothelial cell markers *PODXL*, *KDR* and *ACKR1* compared with those of comparator groups (5% false discovery rate [3]; Figure 1B). Correspondingly, inferred contact [4] between fibroblasts and endothelial cells was significantly enhanced in ICI-IA compared to either SPRA or PsA, and a similar pattern was observed for SIA, albeit less marked (Figure 1C). Other predicted disease group-specific cellular interactions included dendritic cells (DC) with CD8+ T-cells in ICI-IA, and gene set enrichment analysis highlighted relatively enhanced TNF signalling in memory CD8+ T cells (BH-corrected $p=5.21 \times 10^{-5}$), reflecting a pattern in the synovium more generally; indeed, module scoring suggested this significantly exceeded that observed in SPRA ($p=3.02^{49}$; Figure 1D). Spatial corroboration of these and other putative interactions at a transcript level by Xenium analysis is ongoing (Figure 1E).

Conclusion: Our preliminary findings highlight aspects of synovial pathobiology in ICI-IA that are shared to varying degrees with spontaneous arthritis

subtypes. These insights already provide pathobiological support to the current use of TNF inhibitors as a treatment for ICI-IA and could guide the development of targeted treatment strategies for both ICI-IA and SIA in the future.

REFERENCES:

- [1] Donlin LT et al (2018). <https://doi.org/10.1186/s13075-018-1631-y>.
- [2] Krenn V et al (2002). <https://doi.org/10.1078/0344-0338-5710261>.
- [3] Mangiola S et al (2023). <https://doi.org/10.1073/pnas.2203828120>.
- [4] Jin S et al (2024). <https://doi.org/10.1038/s41467-021-21246-9>.
- [5] Korotkevich G et al (2021). <https://doi.org/10.1101/060012>.

Figure 1

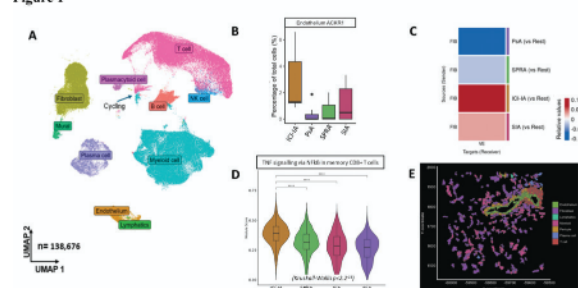


Figure 1. A. UMAP indicating major cell subsets following normalisation, clustering and annotation. B. Proportional abundance of ACR1-expressing endothelial cells across diagnostic categories. C. Heatmap comparing interaction strength between fibroblasts (FIB) and vascular endothelial (VE) cells across arthritides. D. Violin plot comparing gene module scoring of TNF Signalling via NF-κB (MsigDB Hallmark) in CD8+ memory cells across arthritides; module scores calculated using the Seurat ‘AddModuleScore’ function (Kruskal-Wallis with Dunn’s post hoc). E. Synovial spatial transcriptomics displaying perivascular fibroblast association.

Acknowledgements: This work was supported by the Medical Research Council (MR/X02914X/1), GSK and the JGW Patterson Foundation (30015.088/29of20/PA/IXS). Infrastructural support was provided by Versus Arthritis (Research into inflammatory Arthritis Centre; award ref. 22072) and the National Institute of Health and Care Research (NIHR) Newcastle and Birmingham Biomedical Research Centres. The views expressed are those of the authors and not necessarily those of the NIHR or the Department of Health and Social Care.

Disclosure of Interests: Jason D. Turner: None declared, Anthony K. McLean: None declared, Amanda Thomson: None declared, Kristian Williams: None declared, Ioana Nicorescu: None declared, Luke Jones: None declared, George Mercedes: None declared, Ana Teodosio: None declared, Bethany Hunter: None declared, Jack McMurray: None declared, Sebastian Gilbert: None declared, Abbie Degnan: None declared, Dagmar Scheel-Toellner: None declared, Shashank Jariwala GSK shareholder, Current employee of GSK, Qin An GSK shareholder, Current employee of GSK, Ruth Plummer Received payment for delivery of educational talks or chairing educational meetings by AstraZeneca, Novartis, Bayer, MSD and BMS, Received honoraria for working as an IDMC member for Alligator Biosciences, GSK, Onxeo, SOTIO Biotech AG, and AstraZeneca. In the last 3 years received Honoraria for attending advisory boards from Pierre Faber, Bayer, Novartis, BMS, Ellipses, Immunocore, Genmab, Astex Therapeutics, MSD, Nerviano, AmLo, Incyte, Cybrexa, Benevolent AI and Sanofi Aventis, Adam P. Croft: None declared, Lalit Pallan Received payment for speaking for Bristol Myers Squibb, Merck, Received payment for consulting for Bristol Myers Squibb, Iovance, Benjamin A. Fisher Received payment for speaking for Novartis, Servier, Novartis, BMS, Servier, Galapagos, Roche, UCB, Sanofi, Janssen, AZ, Otsuka, Amgen, Kiniksa, Received research funding from Janssen, Servier, Galapagos, Celgene, Novartis, Jonathan Coxhead: None declared, Rafiq Hussain: None declared, Michal Magid-Slav GSK shareholder, Current employee of GSK, David McDonald: None declared, Stefan Siebert Receipt of departmental research grants from GSK, Andrew Filby: None declared, Gary Reynolds: None declared, Amy E Anderson: None declared, Catharien Hilken: None declared, David F. Tough GSK shareholder, Current employee of GSK, Andrew Filer Consultancy for Johnson & Johnson and Sonoma, Institutional grant funding from BMS, Roche, UCB, Nascient, Mestag, GSK, Johnson & Johnson and Synact, Arthur Pratt Funding support from GSK for submitted abstract.

DOI: 10.1136/annrheumdis-2025-eular.A1045

© The Authors 2025. This abstract is an open access article published in Annals of Rheumatic Diseases under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>). Neither EULAR nor the publisher make any representation as to the accuracy of the content. The authors are solely responsible for the content in their abstract including accuracy of the facts, statements, results, conclusion, citing resources etc.